

Cyber Security Tasks & Instructions

- **TASK 1: Professional LinkedIn Announcement**
 - **Objective:** Share your official internship acceptance on your professional network.
 - **Requirements:** Post a high-resolution copy of your official **Internship Offer Letter** on LinkedIn. Craft a compelling caption detailing your learning goals, tag **Progree**, and use hashtags: `#cybersecurity` `#infosec` `#pentesting` `#progree`
- **TASK 2: Host & Network Vulnerability Assessment Lab**
 - **Objective:** Identify structural service configuration flaws within an isolated environment.
 - **Requirements:** Run network reconnaissance utilities (Nmap, Nessus, or Nikto) inside an isolated test environment against target virtual machines. Map active open network ports, audit running software version variations against known CVE vulnerability indexes, and draft a structured 3-page triage prioritization report.
- **TASK 3: Low-Level Network Packet Inspection & Protocol Analysis**
 - **Objective:** Monitor data packet streams to catch network security anomalies or clear-text authentication exploits.
 - **Requirements:** Intercept network data logs using packet capture tools (Wireshark) from a sample threat simulation loop. Analyze TCP handshake protocols, track individual packet stream data layers, isolate unauthorized data exfiltration trends, and identify unencrypted protocol vulnerability configurations (such as FTP or HTTP).
- **TASK 4: Mini Project - Enterprise Threat Modeling Framework & Incident Response Blueprint**
 - **Objective:** Construct a comprehensive security infrastructure architectural blueprint for an enterprise cloud platform.
 - **Requirements:** Model explicit vector threats for a target system architecture utilizing the STRIDE framework. Outline an enterprise network access policy applying Zero-Trust architectures and multi-factor authentication controls. Compile a 4-stage technical incident response lifecycle plan to handle high-risk network compromises (such as active ransomware or database breaches, 5–6 pages).

